

# Federico Melo Barrero

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## EDUCATION

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### Universidad de los Andes

*B.Sc. Systems and Computing Engineering*

Bogotá, Colombia

2020 – 2024

- GPA: 4.92/5.00
- *Summa cum laude*, highest GPA in major in 40 years, highest in Engineering in 16 years

## EXPERIENCE

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### Canals AI

Dec 2024 – Present

Coral Gables, FL, USA · Remote · Full-time

*Software Engineer II*

Jul 2025 – Present

- Owned parsing services for the live voice and inquiries agent systems, as part of a pipeline that processed 20000+ orders daily across 115+ wholesale distributors and 9000+ sales reps
- Built the parsing and review layers of a real-time voice service (WebSocket orchestration, speech-to-text, speaker diarization, concurrency control). Extracted ~21 products per call from live phone calls with 75%+ retained through rep review
- Integrated tools (web search, spec sheet retrieval, ERP lookup, order tracking) into an inquiries AI agent that addressed customer questions on orders, materials, and logistics. Achieved 97% of answers citing a source and 80% rated positive by reps
- Contributed to Canals' shared agent framework (Mastra), adding extended thinking, reasoning effort, image/PDF input, temperature, and tool concurrency. Migrated four Parsing agents off hand-rolled, unbounded loops onto bounded runs with step and token ceilings
- Co-built an autonomous agent (Claude Agent SDK) that polled tickets, debugged parsing failures, and shipped prompt-level fixes for review, re-parsing each order to verify its own fix
- One of 4 engineers on the 10-person Parsing team (~60 engineers total) holding merge approval rights on the main branch.

*Software Engineer I*

Dec 2024 – Jul 2025

- Maintained a parsing pipeline supporting 19 mimetypes with intricate formats (e.g., handwritten, struck-through text, blurry and fragmented photos, cross-references to other files) in a pipeline serving ~60 wholesale distributors processing 4,000+ orders daily
- Abstracted OCR pipeline from single provider to support multiple providers, decoupling the implementation from Textract and enabling format-specific OCR selection, with a cache layer serving ~60% of requests in 0.07s versus 3.6–18.7s uncached
- Led large-scale prompt refactors improving external QA pass rate from 85% to 89%. Involved: model selection, prompt engineering, cache handling, fine-tuning, and regression testing
- Built visual highlighting systems using DOM traversal, coordinate mapping, and string comparison to map parsed products to source documents. Reduced weekly highlighting errors by 2/3
- Joined as the company's 14th engineer.

### Universidad de los Andes

Aug 2025 – Present

*Adjunct Professor, Algorithm Design*

Fall 2026

- Principal instructor for upper-division algorithm design course, delivering lectures and designing exams
- Topics: divide and conquer, dynamic programming, graph algorithms, greedy methods, NP-completeness, approximation and randomized algorithms

*Adjunct Professor, Introduction to Programming*

Fall 2025, Spring 2026

- Principal instructor for introductory undergraduate course in procedural programming using Python
- Topics: programming fundamentals and data-processing libraries Pandas and Matplotlib

## Universidad de los Andes

Bogotá, Colombia · On-site · Part-time

### Full-Stack Developer

Jan 2024 – Dec 2024

- Designed and built full-stack student data analytics system serving 15,000+ students and 600+ faculty, replacing an external platform and saving ~\$100K USD annually
- Built parallelized data processing pipeline (Python, Polars) with complex filtering and aggregations, exposed through a unified REST API (FastAPI)
- Developed dynamic graphing, advanced filtering, report generation, and role-based dashboards in a React/TypeScript web app
- Built early-warning system for at-risk students (academic performance, dropout indicators), alerting counselors on 1,000+ critical cases
- Architected secure cross-VM communication using air-gapping and network isolation techniques to ensure compliance with student data protection requirements

## Caseware International Inc.

Toronto, ON, Canada · Remote · Internship

### Software Developer I

Jan 2024 – Jul 2024

- Awarded June Team Award for impactful contributions as part of Data Analytics team
- Developed Jupyter notebooks for financial and auditing data analysis
- Optimized microservices (Java, TS) performance using AWS S3 and DynamoDB
- Authored technical reports influencing team architecture decisions and best practices

## Universidad de los Andes

Bogotá, Colombia · Hybrid · Part-time

### Backend Developer

Aug 2023 – Dec 2023

- Designed and implemented backend architecture for the Professors' Portfolio system (Java, Spring Boot), a web application for managing faculty academic and administrative information.
- Built persistence, logic, and control layers with CRUD operations, custom logic, and RESTful APIs
- Devised efficient data migration methods for unstructured faculty data
- Collaborated on system architecture documentation and project planning roadmaps

## AWARDS & HONORS

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### Dejar Huella Award

Universidad de los Andes, 2026

*Uniandes internal impact award, for a peer-support platform serving 56,000+ students*

### Award for Outstanding Saber Pro and TyT Scores 2024

Colombian Ministry of National Education, 2025

*Colombia's mandatory national exit exam for graduating university students*

### Summa Cum Laude

Universidad de los Andes, 2025

*Highest undergraduate honor; top 1% cumulative GPA among graduates of the faculty*

### Athletic Career Distinction

Universidad de los Andes, 2025

*Awarded to one graduating student university-wide for leadership and excellence in varsity athletics*

### Ramón de Zubiría Award (2 times)

Universidad de los Andes, 2023, 2024

*Awarded annually to the top cumulative-GPA student in the major*

### Semester Excellence Award (3 times)

Universidad de los Andes, Spring 2021, Fall 2021, Fall 2023

*Awarded each semester to the top-GPA student in the major*

## TECHNICAL EXPERIENCE

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**AI Engineering & LLM Orchestration:** Anthropic / OpenAI / Gemini / Groq APIs, Claude Agent SDK, Mastra, MCP

**Backend & Systems Architecture:** Python, Java, TypeScript, FastAPI, Spring Boot, WebSocket Orchestration, Concurrency Control

**Data Engineering & Infrastructure:** Polars, PostgreSQL, Redis, AWS (RDS, ECS, S3, DynamoDB), Docker, GitHub CI/CD

**Product & Interface:** React, Next.js, Svelte, Plotly, ShadCN / MUI